Software Requirements Specification (SRS)

Smart Food Donation System

1. Introduction

The Smart Food Donation System is a web-based application designed to reduce food wastage and help distribute surplus food to people in need.

The platform connects food donors such as restaurants, hotels, and households with NGOs and volunteers who can collect and distribute the food.

This system ensures that food donations are managed efficiently and delivered before the food expires.

2. General Description

The system provides an online platform where donors can post details about surplus food. NGOs and volunteers can view these posts and coordinate pickup and delivery.

The system consists of four main users: Donor, Volunteer, NGO, and Admin. Donors add food details, volunteers collect and deliver food, NGOs receive and confirm delivery,

and the admin manages and verifies users.

3. Functional Requirements

• User Registration and Login for donors, volunteers, NGOs, and admin.

• Donor can add food donation details (food type, quantity, location, expiry time).

• Volunteer can view available donations and accept pickup requests.

• NGO can confirm receiving the food donation.

• Admin can manage users, verify NGOs, and monitor all system activities.

• Notification system for donation status updates.

4. Interface Requirements

User Interface:

- HTML, CSS, JavaScript, Bootstrap.

Example Interfaces:

• Login Page

• Registration Page

• Donor Dashboard

• Volunteer Dashboard

• NGO Dashboard

• Admin Dashboard

Software Interface:

- Chrome/Firefox browser, XAMPP server, MySQL database.

5. Performance Requirements

The system should respond quickly to user requests and support multiple users simultaneously.

Food donation information should update in real time to avoid delays.

6. Design Constraints

• Developed using PHP, MySQL, HTML, CSS, JavaScript.

• Runs on XAMPP server.

• Must include user authentication and basic security.

7. Non-Functional Attributes

Security – User authentication and admin verification.

Reliability – System should run continuously without failure.

Usability – Easy-to-use interface.

Scalability – Future expansion such as mobile app support.

8. Preliminary Schedule and Budget

– Requirement analysis

– System design

– Frontend development

– Backend development

– Testing

– Documentation

Budget – Minimal because tools (HTML, CSS, PHP, MySQL, XAMPP) are open source.

9. Appendices

Technologies Used:

Frontend – HTML, CSS, JavaScript, Bootstrap

Backend – PHP

Database – MySQL

Server – Apache (XAMPP)

Optional APIs:

Google Maps API

Email/SMS API

10. Uses of SRS Document

Helps developers understand system requirements.

Acts as a development guideline.

Helps testers verify system functionality.

Useful for future upgrades.

11. FAQs on SRS Format

What is SRS?

It is a document describing the requirements of a software system.

Why is SRS important?

It ensures developers understand system requirements before development.

Who uses SRS?

Developers, testers, project managers, and stakeholders.

12. Conclusion

The Smart Food Donation System helps reduce food wastage and feed people in need by connecting donors, volunteers, and NGOs through a digital platform.